

LAB EXPERIMENT RECORD

Set 9: Online_Learning_Activity

Department	CSE-Ds
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	03 October 2026
Day	Saturday

AIM

To analyze online learning activity data using R by creating a histogram, boxplot, scatter plot, and line chart with moving average smoothing, and to design a mini interactive dashboard for exploring learning performance.

DATASET: Online_Learning_Activity.csv

Student ID	Gender	Age	Course	Study Time (hrs)	Videos Watched	Quiz Score	Login Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08

Student ID	Gender	Age	Course	Study Time (hrs)	Videos Watched	Quiz Score	Login Date
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

PROGRAM 1: Histogram and Boxplot of Quiz Score

Aim

To import the online learning dataset and create a histogram and boxplot to analyze student quiz performance.

Program

R

```
data <- read.csv("Online_Learning_Activity.csv")
```

```
hist(data$Quiz_Score,
```

```
  xlab = "Quiz Score",
```

```
  ylab = "Number of Students",
```

```
  main = "Distribution of Quiz Scores")
```

```
boxplot(Quiz_Score ~ Course,
```

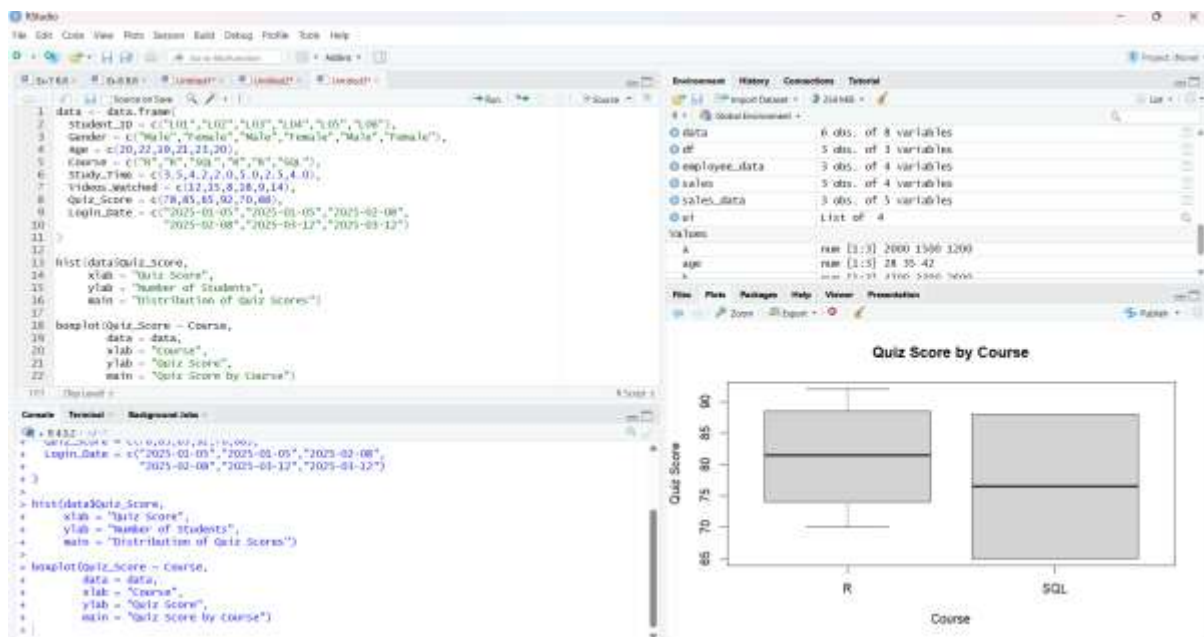
```
  data = data,
```

```
  xlab = "Course",
```

```
  ylab = "Quiz Score",
```

```
  main = "Quiz Score by Course")
```

Output



The histogram shows the distribution of quiz scores, and the boxplot compares quiz scores between R and SQL courses.

Observation

Quiz scores range from 65 to 92. Students in the R course generally have good performance, while the SQL course has a wider difference in scores.

PROGRAM 2: Scatter Plot with Bubble Size

Aim

To create a scatter plot between study time and quiz score, where the bubble size represents the number of videos watched.

Program

R

```
data <- read.csv("Online_Learning_Activity.csv")
```

```
plot(data$Study_Time..hrs.,
```

```
      data$Quiz_Score,
```

```
      cex = data$Videos_Watched / 5,
```

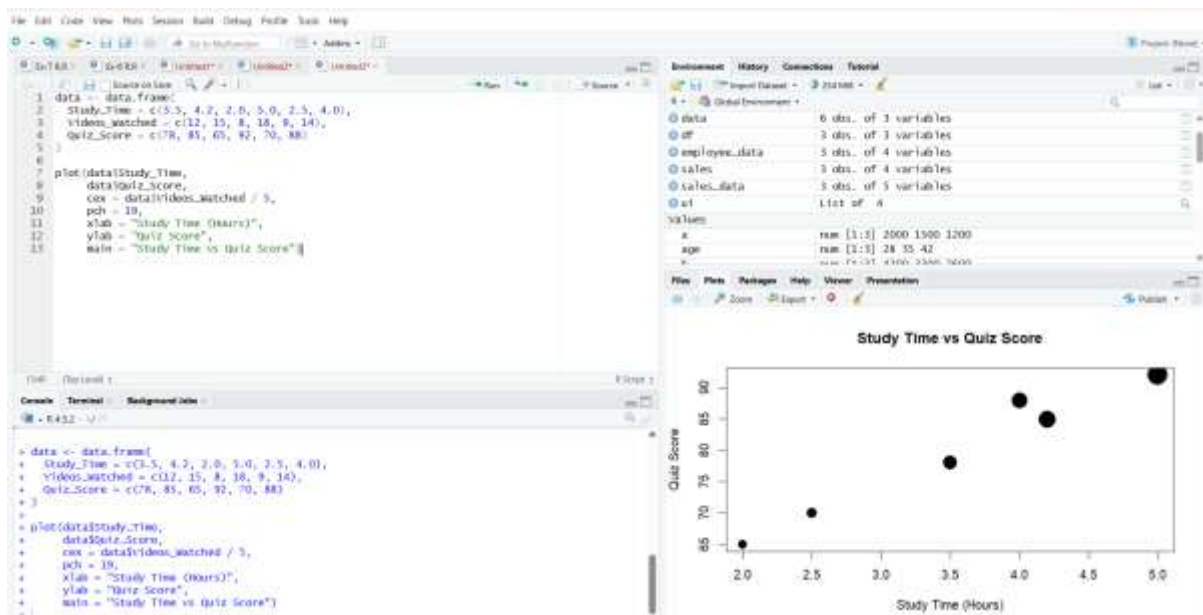
```
      pch = 19,
```

```
      xlab = "Study Time (Hours)",
```

```
      ylab = "Quiz Score",
```

```
main = "Study Time vs Quiz Score")
```

Output



A scatter plot showing the relationship between study time and quiz score. Larger bubbles represent students who watched more videos.

Observation

Students who spend more time studying generally achieve higher quiz scores. Students who watch more videos also tend to have better learning performance.

PROGRAM 3: Monthly Average Quiz Score with Moving Average

Aim

To convert the login date into Date format, calculate the average quiz score for each month, and display the trend using a line chart with moving average smoothing.

Program

R

```
data <- read.csv("Online_Learning_Activity.csv")
```

```
data$Login_Date <- as.Date(data$Login_Date)
```

```
data$Month <- format(data$Login_Date, "%Y-%m")
```

```
monthly_score <- aggregate(Quiz_Score ~ Month,
```

```
    data = data,
```

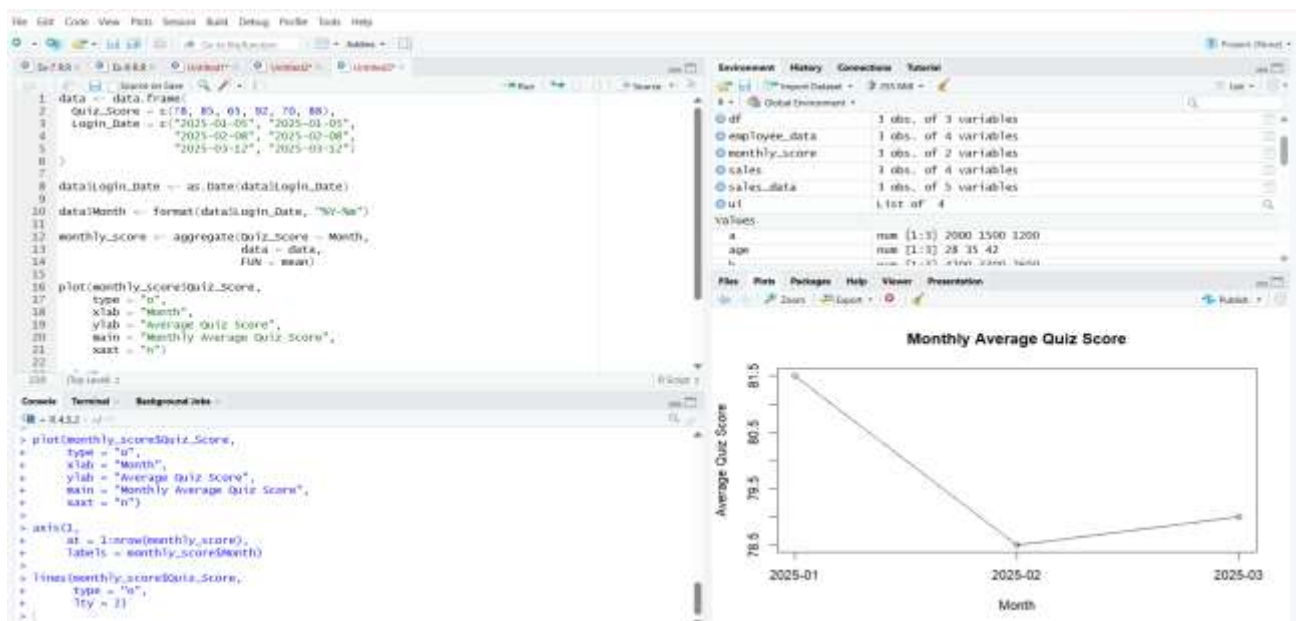
```
    FUN = mean)
```

```
monthly_score$Moving_Average <- monthly_score$Quiz_Score
```

```
plot(monthly_score$Quiz_Score,  
     type = "o",  
     xlab = "Month",  
     ylab = "Average Quiz Score",  
     main = "Monthly Average Quiz Score",  
     xaxt = "n")
```

```
axis(1, at = 1:nrow(monthly_score),  
     labels = monthly_score$Month)  
lines(monthly_score$Moving_Average,  
      type = "o",  
      lty = 2)
```

Output



A line chart displaying the monthly average quiz scores and the moving average trend.

Observation

The average quiz score changes across the months. The trend helps to understand changes in student performance over time.

PROGRAM 4: Mini Interactive Dashboard

Aim

To create a mini interactive dashboard containing a bar chart of average quiz score by course and a scatter plot of study time versus quiz score with filters for gender and course.

Program

R

```
library(shiny)

data <- read.csv("Online_Learning_Activity.csv")

ui <- fluidPage(

  titlePanel("Online Learning Activity Dashboard"),

  sidebarLayout(

    sidebarPanel(

      selectInput("gender",

                  "Select Gender:",

                  choices = c("All",

                              unique(data$Gender))),

      selectInput("course",

                  "Select Course:",

                  choices = c("All",

                              unique(data$Course)))

    ),

    mainPanel(

      plotOutput("barChart"),

      plotOutput("scatterChart")

    )

  )

)
```

```

server <- function(input, output) {

  filtered_data <- reactive({

    result <- data

    if (input$gender != "All") {

      result <- result[result$Gender == input$gender, ]

    }

    if (input$course != "All") {

      result <- result[result$Course == input$course, ]

    }

    result

  })

  output$barChart <- renderPlot({

    result <- filtered_data()

    avg_score <- aggregate(

      Quiz_Score ~ Course,

      data = result,

      FUN = mean

    )

    barplot(avg_score$Quiz_Score,

      names.arg = avg_score$Course,

      xlab = "Course",

      ylab = "Average Quiz Score",

      main = "Average Quiz Score by Course")

  })

  output$scatterChart <- renderPlot({

    result <- filtered_data()

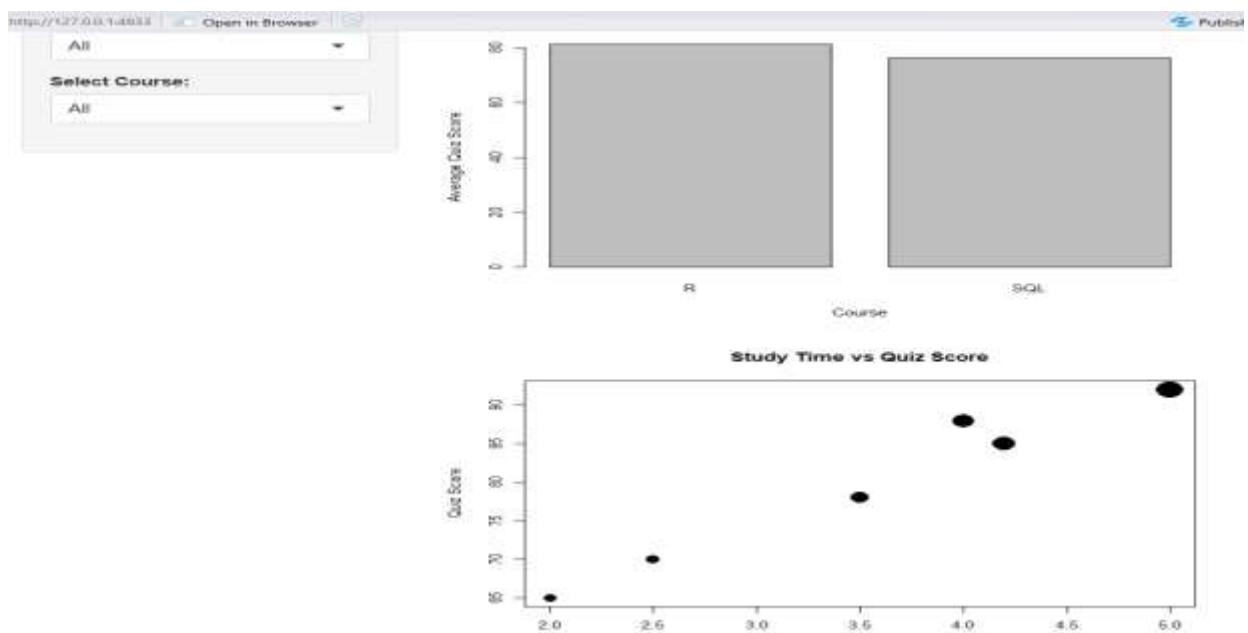
```

```

plot(result$Study_Time..hrs.,
      result$Quiz_Score,
      cex = result$Videos_Watched / 5,
      pch = 19,
      xlab = "Study Time (Hours)",
      ylab = "Quiz Score",
      main = "Study Time vs Quiz Score")
})
}
shinyApp(ui = ui, server = server)

```

Output



Observation

The dashboard allows users to select gender and course and interactively explore the relationship between study time, videos watched, and quiz performance.